Computational Statistical Techniques: Jackknife and Bootstrap with Applications in Correlations, PCA, and Hypothesis Testing

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Bootstrap Methods for Simple and Partial Correlation

Let $(X_1, Y_1, Z_1), \ldots, (X_n, Y_n, Z_n) \sim F$. The partial correlation coefficient of X, Y given Z is given by:

$$\theta = -\frac{\Omega_{12}}{\sqrt{\Omega_{11} \cdot \Omega_{22}}}$$

where $\Omega = \Sigma^{-1}$ and Σ is the variance–covariance matrix of $W = (X, Y, Z)$. The partial correlation coefficient θ measures the linear dependence between X and Y when Z is held constant (for any fixed value of Z).

A) Using the data from the file Games.txt and $B = 1000$ bootstrap samples, we construct all 95% confidence intervals for the correlation coefficient of X, Y. The resulting intervals are as follows:

Confidence Intervals		
	LCL	UCL
Basic Bootstrap	0.9832292	0.9939088
Percentile Bootstrap	0.980617	0.9912967
BCa Bootstrap	0.9805669	0.9912447
Bootstrap-t	0.9811067	0.9918753

Table 1: 95% confidence intervals for the correlation coefficient of X, Y using four bootstrap methods.

In all confidence intervals, we observe that the correlation coefficient is very close to 1. All intervals are near 1 and are very narrow, indicating an almost perfect positive linear relationship between X and Y. We also note that all four methods yield very similar intervals. None of the intervals contain 0, meaning the correlation is statistically significant. The agreement among the four different bootstrap methods strengthens our confidence in the estimate.

B) Below we will construct all 95% confidence intervals for the partial correlation coefficient θ. The intervals we obtain are as follows:

Confidence Intervals		
	LCL	UCL
Basic Bootstrap	-0.214115	0.1456985
Percentile Bootstrap	-0.2447511	0.1150624
BCa Bootstrap	-0.2336297	0.1214474
Bootstrap-t	-0.2125329	0.1697444

Table 2: 95% confidence intervals for the partial correlation coefficient θ of X, Y given Z, using four bootstrap methods.

Based on the results, the seemingly perfect correlation between X and Y proves to be entirely spurious when the effect of the third factor Z is taken into account. The partial correlation X, Y | Z ranges between -0.21 and 0.17, with all confidence intervals containing zero, indicating that there is no statistically significant direct relationship between the two variables. This dramatic change reveals that Z acts as a common confounding factor: both X and Y are likely influenced simultaneously by Z, creating the illusion of a strong linear connection between them. Essentially, the "relationship"

we initially observed was a statistical artifact; the true structure of the data emerges only when we isolate the effect of the hidden factor Z.

Bootstrap and Jackknife Methods for PCA and Hypothesis Testing

Consider the dataset BodyMeasurements.txt, which contains the results of 5 body measurements from 100 individuals. Let Σ be the variance-covariance matrix of the data and let its eigenvalues be $\lambda_1 > \lambda_2 > \lambda_3 > \lambda_4 > \lambda_5$. Then,

$$\theta = \frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5}$$

is the quantity to be estimated.

A) We will use the jackknife estimator to estimate the bias and the standard error of the estimator of θ. Below we obtain the values we are looking for.

JackKnife		
Estimate of θ	Standard Error	Bias
0.8754452	0.01718302	-0.0001075565

Table 3: Jackknife estimate of θ, standard error, and bias.

The jackknife results show that the estimate of the proportion of variance explained by the first principal axis (θ ≈ 0.875) is extremely reliable and almost unbiased. The value θ = 0.875 indicates a very strong linear correlation among the five body measurement variables, with the first axis explaining 87.5% of the total variance. The small standard error (0.017) demonstrates high precision in the estimate, while the nearly zero bias (-0.0001) confirms that the estimator is practically unbiased for a sample of 100 individuals.

B) We will use B = 1000 bootstrap samples to construct the 4 basic 95% confidence intervals for the quantity θ.

Confidence Intervals		
	LCL	UCL
Basic Bootstrap	0.8451209	0.9122948
Percentile Bootstrap	0.8383804	0.9055543
BCa Bootstrap	0.8314404	0.9021037
Bootstrap-t	0.8354003	0.9049666

Table 4: 95% confidence intervals for θ using four bootstrap methods

The results of the four different bootstrap methods provide consistent and reliable estimates for the quantity θ, which represents the proportion of the total variance explained by the first principal axis. Based on the confidence intervals, and with 95% statistical certainty, we can conclude that the first principal axis explains between 83% and 91% of the variance of the five body measurement variables. The agreement among all methods, combined with the narrow range of the intervals, indicates that this proportion is estimated with high precision.

C) We will use B = 1000 bootstrap samples to test the hypothesis

$$H_0: \mu_1 = \frac{7}{17}\mu_5 \quad VS \quad H_1: \mu_1 \neq \frac{7}{17}\mu_5$$

at α = 5%, where μ_1 and μ_5 are the means of the first and fifth columns, respectively. Applying this procedure, we obtain a p-value = 0.036, from which we conclude that we reject the null hypothesis and that there is insufficient evidence that $\mu_1 = \frac{7}{17}\mu_5$.

D) Let variable Z represent the second column of the data. We want to estimate the probability P(Z > 55). Specifically, we estimate θ using the simple empirical probability estimator and construct the four basic 98% confidence intervals using Bootstrap with B = 1000 replications.

Confidence Intervals		
	LCL	UCL
Basic Bootstrap	0.05	0.21
Percentile Bootstrap	0.05	0.21
BCa Bootstrap	0.05	0.20
Bootstrap-t	0.06	0.24

Table 5: 98% confidence intervals for the probability P(Z > 55) using four bootstrap methods.

The results of the four different bootstrap methods provide consistent and reliable estimates for the probability θ = P(Z > 55). Based on the confidence intervals and with 98% statistical certainty, we can conclude that the true proportion of individuals with Z > 55 in the population lies between 5% and 21%. The agreement among the first three methods (basic, percentile, BCa), combined with the relatively narrow range of the intervals, indicates that this probability is estimated with good precision. The wider interval provided by the bootstrap-t method is likely due to the method's sensitivity to non-normality of the estimator and the use of double bootstrap, which reinforces the selection of the consistent results from the other three methods.

E) Instead of equal-probability sampling for testing a mean, we consider selecting bootstrap samples with probabilities $p_1, \ldots, p_n$ that:

- Satisfy the constraint: $\sum p_i x_i = \mu$ (mean under the null hypothesis)
- Minimize the Kullback–Leibler divergence

Using the data from the first column, estimate the p-value for testing the hypothesis

$$H_0: \mu_1 = 52 \quad VS \quad H_1: \mu_1 \neq 52$$

at α = 5%, for:

- Classical t-test
- Ordinary Bootstrap
- KL Bootstrap

Hypothesis Testing	
Test Method	p-value
t-test	0.047
Bootstrap	0.052
Bootstrap-KL	0.040

Table 6: Comparison of p-values for testing the mean using t-test, Bootstrap, and Bootstrap-KL methods.

The results of the three methods yield p-values of 0.047 (t-test), 0.052 (ordinary bootstrap), and 0.040 (KL bootstrap). At the 5% significance level, the t-test and KL bootstrap lead to rejection of the null hypothesis $H_0: \mu_1 = 52$, while the ordinary bootstrap gives a marginally non-significant result (p-value = 0.052 > 0.05). This slight discrepancy is likely due to sampling variability. Nevertheless, the overall evidence supports rejecting the null hypothesis, indicating that the mean of the first column is statistically significantly different from 52.